The Itô stochastic differential equation [1] is one of the foundations in finance:

$$dX_t = b(X_t, t)\,dt + \sigma(X_t, t)\,dW_t.$$

In stochastic calculus Markov [2] play a predominant role and under certain conditions Itô can be viewed as such. By the Markov property we can employ the Chapman-Kolmogorov equation [2]:

$$K_s, t(x, A) = \int K_u, t(y, A), K_s, u(x, dy).$$

We can construct a category $\mathcal{C}$, Lean code here, whose objects are probabilistic state spaces and whose morphisms are Markov kernels. Denoting the transition kernel by $K_{s,t}$:

$$K_{s,t}(x, A) = \mathbb{P}(X_t \in A \mid X_s = x).$$

The Chapman-Kolmogorov equation becomes:

$$K_{s,t} = K_{s,u} \circ K_{u,t}.$$

References

[1] Emanuela Gianin, Esercizi di finanza matematica, 2007.

[2] Andrea Pascucci, Probability theory 2, 2024.